A Bayesian factor analytic approach for handling missing data in spatial regressions models

Indira Puteri Kinasih[1,2], Haziq Jamil[1], Elvynna Leong[1]

ARTICLE HISTORY
Compiled December 21, 2025

[1] Mathematical Sciences, Faculty of Science, Universiti Brunei Darussalam, Bandar Seri Begawan, Brunei
[2] Mathematics Education, Faculty of Education, Universitas Islam Negeri Mataram,

ABSTRACT
Missing data remains a critical challenge in spatial modelling, particularly in real-world datasets where covariate information is often incomplete. This study introduces an integrated Bayesian framework that combines a factor analysis-based imputation method with a multilevel Conditional Autoregressive (mlvCAR) model to improve inference in hierarchical spatial datasets with missing covariates. By using latent factor structures, the method reduces the dimensionality of the covariate space, simplifying the imputation process without explicitly modeling the missingness mechanism. The spatial dependence is preserved through the inclusion of a Leroux CAR prior, enabling the imputed values to remain spatially coherent. A simulation study evaluates the approach under both Missing Completely at Random (MCAR) and Missing at Random (MAR) conditions across varying levels of missingness. Results demonstrate that the proposed framework achieves lower bias and more stable parameter estimates compared to complete-case analysis. An application to house price data in Brunei-Muara further highlights the practical value of the method, especially in settings with limited or incomplete covariate information. This balance of parsimony and spatial sensitivity makes the approach well-suited for real-world spatial data challenges.

KEYWORDS
missing data; factor analysis; spatial regression; conditionally autoregressive model; property prices

1. Introduction

Spatial regression models, particularly Conditionally Autoregressive (CAR) models, are commonly employed in spatial regression analysis to examine spatially structured data across various fields. These models have been widely applied in several study areas (Donegan 2021; D. Lee 2013; D. Lee, Rushworth, and Napier 2018; Lawson et al. 2015; Freni-Sterrantino, Ventrucci, and Rue 2017; Kinasih, Jamil, and Leong 2025). However,

CONTACT: Indira Puteri Kinasih. Email: indiraputeri@uinmataram.ac.id. Haziq Jamil. Email: haziq.jamil@ubd.edu.bn. Elvynna Leong. Email: elvynna.leong@ubd.edu.bn.

a persistent challenge in applying these models arises from missing data in covariates, which is a common issue when working with real-world datasets, see for instance Liu et al. (2023), Ma, Hu, and Chen (2020), and W. Wang and Lee (2013). Missing data in spatial datasets often follow non-random patterns, such as spatial or temporal clustering, making conventional imputation techniques ineffective, especially when large portions of the spatial domain are affected (Arnone, Sangalli, and Vicini 2021). Meanwhile, CAR models, like other spatial regression models, typically assume complete covariates data, and ignoring or naively handling missing values can lead to biased estimates and unreliable inferences. This challenge limits the broader application of CAR models in spatial analysis, see Figure 1.

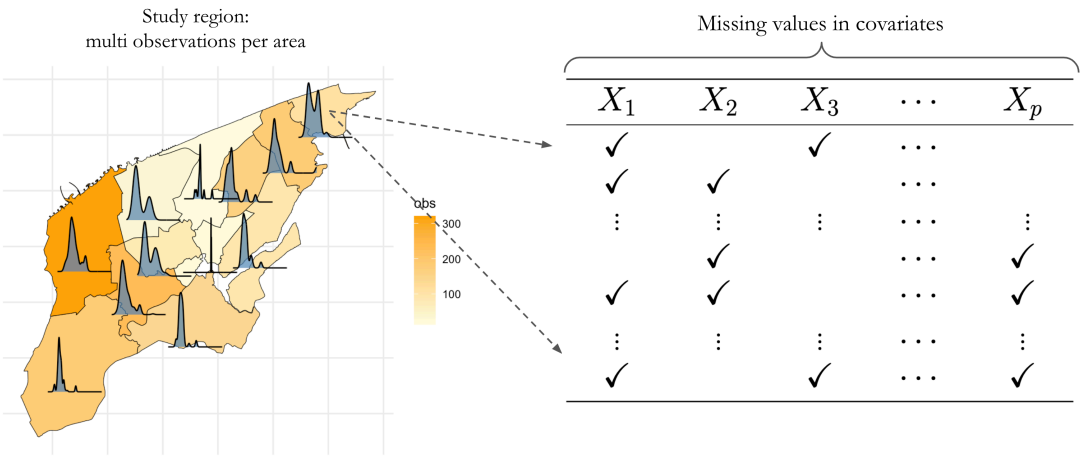

Figure 1. problem arise in spatial regression modelling: multiobservation in each area unit, different distribution (showed by the density plots) and missing values in the covariates

Several studies have addressed missing data in spatial models using conventional imputation techniques, such as spatial mean imputation, nearest neighbour interpolation, or multiple imputation methods, for example Gong et al. (2023), Li et al. (2020), and Baker, White, and Mengersen (2014). While some of these approaches incorporate spatial structure, they often do so in a simplistic manner, failing to fully capture complex spatial dependencies and localised variations in the data. Moreover, many traditional strategies adopt a two-stage process by first imputing the missing values, then fitting a spatial model such as a Conditional Autoregressive (CAR) model to the completed dataset (Liu et al. 2023).

Although Bayesian frameworks offer the flexibility to jointly model missing data within the overall model structure, they typically require explicit assumptions about inter-variable relationships (Skarstein, Martino, and Muff 2023; Z. Wang and Tang 2020), for instance, modelling the number of bedrooms as a function of floor area and number of bathrooms. However, specifying such conditional dependencies can be challenging, especially when the true relationships among covariates are not known a priori. This process often involves extensive model testing and variable selection, increasing both the complexity and the computational burden of model development (Du et al. 2021).

In contrast, our proposed approach integrates factor analysis-based imputation directly within the CAR framework in a streamlined and coherent manner. This enables the estimation of missing values while preserving spatial dependencies, without requiring extensive conditional modelling of covariates. Rather than treating imputation as a sep-

arate preprocessing step, we adopt a fully Bayesian strategy that jointly models missing data and spatial effects within a unified hierarchical framework. Our method leverages the underlying correlation structure among covariates, using posterior sampling to recover missing values while respecting spatial grouping and inter-variable relationships. By embedding the imputation process within the model estimation itself, spatial patterns are retained, uncertainty is properly propagated, and statistical inference remains internally consistent. A notable advantage of using factor analysis-based imputation is its ability to capture the latent correlation structure among covariates without needing to explicitly specify conditional relationships. This is particularly valuable in situations where substantive domain knowledge is limited, or exhaustive model comparisons are impractical. Furthermore, the inclusion of spatial random effects allows the model to account for spatial heterogeneity, ensuring that each area is uniquely characterised. While this study focuses on house price modelling, the proposed framework is broadly applicable to other domains and can be flexibly adapted to various Bayesian spatial models beyond CAR. Overall, this research contributes a principled and robust methodology for handling missing data in spatial econometric models, offering a scalable solution for real-world datasets with complex structures and incomplete observations.

The remainder of this paper is organised as follows. We begin with a review of relevant literature, followed by an overview of the proposed modelling framework, including the multilevel CAR model, factor analysis components, and the integration of missing data handling. The Brunei-Muara district is then introduced as the study region. This is followed by a detailed description of the experimental design, comprising both simulation and empirical applications. The results section presents key findings from the imputation experiments and the Brunei-Muara house price case study. The paper concludes with a discussion of the implications and potential directions for future research.

2. Related works

Handling missing data in spatial and spatio-temporal contexts has been the focus of extensive research, especially given the growing availability of complex urban and property-related datasets. Conventional imputation methods, such as spatial mean imputation and nearest neighbour interpolation, are widely used due to their simplicity and intuitive reliance on spatial proximity (Miao et al. 2023). However, these methods often fail to capture more intricate spatial or temporal dependencies and can introduce bias or reduce statistical power in downstream analyses.

Multiple imputation (MI) methods have been widely recognized as robust approaches for handling missing data, as they generate multiple plausible datasets and combine results to account for uncertainty (Beesley et al. 2021; Demirtas 2018). While MI techniques are highly effective, they often face challenges when applied to high-dimensional data or datasets with strong spatial dependence (Brini and Heuvel 2023). Additionally, implementing MI in practice presents several difficulties, including managing multilevel data structures, addressing systematic missingness patterns, handling high-dimensional and non-normally distributed variables, ensuring valid inference through bootstrapping, and improving documentation and sensitivity analyses (Grund, Lüdtke, and Robitzsch 2018; Nguyen, Carlin, and Lee 2021; Beesley et al. 2021). Despite these advancements, MI methods specifically designed to accommodate spatial data remain relatively rare, highlighting a gap in existing methodologies.

Recent developments have focused on more advanced approaches that leverage spatial correlation and machine learning. For example, Gong et al. (2023) proposed a spatial multi-kernel learning approach for imputing missing values in urban statistical data, demonstrating improved accuracy by explicitly incorporating both regional similarities and geographic positions. In the transportation domain, Li et al. (2020) developed a spatiotemporal imputation method using tensor decomposition and graph-based modelling to handle complicated missing patterns in traffic data. Brini and Heuvel (2023) addressed missing data challenges in high-dimensional settings, presenting methods that improve inference when dealing with large numbers of covariates. Bertsimas, Pawlowski, and Zhuo (2017) introduced an optimization-based approach (opt.impute) that formulates imputation as a predictive optimization problem. This method is capable of handling mixed data types and has shown superior performance compared to classical imputation techniques.

In the econometrics and statistical modelling literature, Liu et al. (2023) address missing response issues in spatial dynamic panel data (SDPD) models by proposing a weighted maximum likelihood estimator (WMLE) under a missing at random (MAR) assumption. They also introduce an imputation method that leverages spatial and temporal dependencies along with covariates. Their approach is validated through simulations and a real data application. Additionally, optimization-based frameworks such as opt.impute have been introduced to handle mixed data types under complex missingness patterns using formal optimization techniques (Miao et al. 2023). Another advanced approach involves deep learning, as demonstrated by Zhang et al. (2018), who developed a spatial–temporal–spectral deep convolutional neural network for reconstructing missing data in remote sensing images, effectively capturing dependencies across multiple dimensions.

Zahmatkesh and Mohammadzadeh (2020) introduce a Bayesian hierarchical approach for handling missing spatial data under a non-ignorable missingness (MNAR) setting. Their model employs a Conditional Autoregressive (CAR) prior to capture spatial dependencies while explicitly modelling the missing data mechanism. This approach is particularly useful for datasets where the probability of missing values depends on the observed data itself. However, while their model effectively accounts for spatial autocorrelation, it does not explicitly address latent structures within the data. In contrast, our work introduces a Bayesian factor analytic approach, which not only accounts for spatial dependence via a Leroux CAR prior but also captures underlying latent patterns that influence missingness and spatial regression modelling. This distinction allows our method to handle high-dimensional dependencies more efficiently while addressing both missing data uncertainty and spatial effects simultaneously.

3. Models

Property price modelling involves analyzing and understanding various factors that influence property values, be it intrinsic features (such as property characteristics) or extrinsic features (such as neighbourhood characteristics). Let Y denote the price of a property, and let $\mathbf{X} = \{X_1, \dots, X_p\}$ represent its characteristics such as size, property type, number of bedrooms, bathrooms, and so on. In what follows, we describe the *multilevel conditionally autoregressive (mlvCAR) model* as a tool to model property prices in a spatially structured manner. Furthermore, we will employ a *Bayesian factor-analytic* approach to model the property characteristics, and propose an imputation

method to handle missing data.

3.1. *Multilevel CAR model*

Suppose one has, for analysis, non-overlapping spatial areas indexed by $i = 1, \dots, M$. Within each spatial area, there are n_i observations, indexed by $j = 1, \dots, n_i$. The multilevel CAR model may be specified as

$$Y_{ij} = \beta_0 + \sum_{k=1}^{p} \beta_k X_{ijk} + \phi_i + \epsilon_{ij}, \tag{1}$$

where ϵ_{ij} represents the error term, β_0 is the intercept, β_k are the regression coefficients for the k-th covariate X_{ijk}, and ϕ_i is a random effect associated with the i-th area.

A fair assumption when dealing with multilevel areal data is to expect heterogeneity among the areas and homogeneity within the areas–a common phenomenon often referred to as the "between" and the "within" effects. The goal of the random effect ϕ_i is to account for the spatial correlation between the observations within the same area. The spatial autocorrelation is enforced by a conditional autoregressive (CAR) prior on the random effects $\phi = (\phi_1, \dots, \phi_M)^\top$. Specifically, ϕ is assigned the Leroux CAR prior

$$\phi \sim \mathcal{N}(\mathbf{0}, \tau^{-1}(\mathbf{D} - \rho\mathbf{W})^{-1}). \tag{2}$$

Here, $\mathbf{W}$ is a symmetric matrix has represents the adjacency matrix that encodes the neighbourhood structure; it has elements $w_{im} = 1$ if areas i and m are neighbours, and 0 otherwise. Note that the diagonals of $\mathbf{W}$ are zero, since a region is not considered a neighbour of itself. Further, $\mathbf{D}$ is a diagonal matrix with elements $d_{ii} = \sum_m w_{im}$, capturing the total spatial connections for each area. The remaining parameters are τ_ϕ, which controls the spatial precision and influences the smoothness of the spatial effect, and ρ, which governs the strength of spatial dependence, determining how much an area's effect is influenced by its neighbours.

Evidently, the Leroux CAR prior balances between spatial dependence and independent random effects. This may be seen more clearly when the Leroux CAR prior is expressed in terms of the conditional distribution of ϕ_i given the other random effects ϕ_{-i}:

$$\phi_i \mid \phi_{-i} \sim \mathcal{N}\left(\frac{\rho \sum_{m=1}^{M} w_{im}\phi_m}{A}, \frac{\tau^{-1}}{A}\right) \tag{3}$$

where $A = \rho \sum_{m=1}^{M} w_{im} + 1 - \rho$. When $\rho = 1$, the prior reduces to the intrinsic CAR prior as described by Besag, York, and Mollié (1991), implying full spatial dependency. Conversely, when $\rho = 0$, the spatial structure is removed, and the model behaves as an independent random effects model. The Leroux conditional autoregressive (CAR) prior may later be denoted as $\phi_i \sim \text{CAR}(\mathbf{W}, \rho, \tau_\phi)$

3.2. *Factor analysis models*

Factor analysis is a classical technique for modelling the covariance (or correlation) structure among a set of observed variables $X_1, \dots, X_p$ by positing the existence of latent factors that influence the observed variables. In the simplest case of a one-factor model, we assume that a single latent variable η captures the common variance in $\mathbf{X}$, while the remaining variability captured by δ is idiosyncratic. The model is written

$$X_k = \lambda_k \eta + \delta_k, \quad k = 1, \dots, p, \tag{4}$$

with $\eta \sim N(0,1)$ and $\delta_k \sim N(0, \theta_k)$. The factor loadings λ_k represent the strength of the relationship between the latent variable and the observed variables, while δ_k are independent unique factors with variances ψ_k. With this formulation, the model-implied covariance structure of the observed variables is given by

$$\mathrm{Cov}(\mathbf{X}) = \Lambda\Lambda^\top + \Theta, \tag{5}$$

where Λ is the $p \times 1$ vector of factor loadings and $\Theta = \mathrm{diag}(\theta_1, \dots, \theta_p)$ is a diagonal matrix of unique variances. This parsimonious representation not only facilitates interpretation of the underlying structure but also yields efficient estimates when p is large relative to the sample size.

In many applied settings such as cross-cultural studies, multi-site clinical trials, or longitudinal surveys, data are usually collected in G distinct groups. In such contexts, it is generally unrealistic to assume common factor loadings (λ) and residual variances in across all groups. Consequently, a separate confirmatory factor analysis (CFA) is often fitted for each group. Let $\mathbf{X}^{(g)} = \{X_1^{(g)}, \dots, X_p^{(g)}\}$ denote the variables in group g. A naïve "group-wise" factor analysis estimates separate loading matrices for each group, like this:

$$X_k^{(g)} = \lambda_k^{(g)} \eta^{(g)} + \delta_k^{(g)}, \quad k = 1, \dots, p. \tag{6}$$

However, this approach is problematic when the number of groups is large. It introduces a substantial number of parameters, suffers from identifiability issues, and ignores the structural relationships between groups, such as spatial dependence. Moreover, due to scale identifiability constraints in factor analysis, one cannot freely estimate both the variance of $\eta^{(g)}$ and the group-specific loadings $\lambda_k^{(g)}$ without imposing additional constraints.

could potentially

Furthermore, it ignores…

This is not worth mentioning. It is true regardless of group or no group analysis

3.2.1. *Idea for spatial FA-based imputation*

This section is not describing imputation

A more satisfactory solution for problems above, borrows strength across groups via a random–effects formulation, allowing both the latent factor (η) and the factor loadings (λ) to vary across areas in a structured way. In some literatures, this is known as

measurement non-invariance (Vermunt and Magidson 2021; Lai 2021; Leitgöb et al. 2022). For consistency, we adopt the same indexing notation for individuals and groups as in Equation 1, where $i = 1, \dots, M$ indexes group (not g) and $j = 1, \dots, n_i$ indexes individuals within group i. First, let

$$\eta^{(i)} = \phi_i + \zeta, \tag{7}$$

wrong, no i subscript

where $\phi_i \sim N(0, \sigma_i^2)$ is the group-level variation and $\zeta \sim N(0, 1)$ controls individual-level variation around it. This decomposes the latent factor into a random effects component and an individual-specific deviation. Note that for an individual j in group i, we have $\eta_j^{(i)} = \phi_i + \zeta_j$ which mean that now we only estimate 1 parameter for group effect, instead of $i = 1, \dots, M$ parameters. Here, ϕ_i is a random effect rather than fixed effect.

Seems redundant to say it again? The sentence that follows is true regardless

effectively treating phi_i as a random effect rather than a fixed effect.

Further, to allow the loading structure to vary across space, we specify for each indicator $k = 1, \dots, p$:

$$\lambda_k^{(i)} = \lambda_k + \kappa_k \phi_i \tag{8}$$

where λ_k is the global loading and κ_k controls the strength of spatial perturbation. This allows loadings to differ across areas without introducing a separate free parameter for each area–indicator pair, only by estimating additional $\kappa_1, \dots, \kappa_p$. A nice property of this spesification is that when $\kappa_k = 0$, the corresponding loading reduces to the global value λ_k, implying no group effects on that covariates (measurement-invariant).

i.e. we have induced correlations between loadings in the same group, but not in different groups. This can be interpreted as ….

when kappa is not 0, then we have cov(lambda_k^(i), lambda_l^(i)) = kappa_k kappa_l sigma^2 \neq 0

Finally, rather than treating the group effects ϕ_i as independent ("vanilla") random effects, we later impose a CAR prior as stated in Equation 3. This choice reflects the fact that in our case, the groups correspond to geographical regions such as district or sub-district, where neighbouring regions are likely to exhibit similar latent characteristics. The CAR prior therefore enables the factor structure to vary smoothly across space while borrowing strength across adjacent regions.

I don't think this is a true statement.
"smooth" usually implies some kind of fine spatial mesh.
we only have small numnber of (contiguous) areas/polygons

3.3. *A unified spatial factor analytic mlvCAR model for missing data*

Under a Bayesian framework, the factor-analytic measurement model and the multilevel CAR (mlvCAR) outcome model are written as a single hierarchy (Figure 2). We assume that missing covariate values arise under a Missing at Random (MAR) mechanism. Under MAR, the missingness process does not contribute to the likelihood, and inference can be based solely on the observed data model.

Within this framework, missing covariates are treated as latent variables and are updated as part of the posterior sampling scheme. Bayesian estimation therefore allows us to sample from the joint posterior state space, simultaneously imputing missing values while estimating all model parameters. This joint modelling strategy ensures that uncertainty arising from missing data is naturally propagated through to parameter estimates and predictions.

A latent factor η captures the shared variance among covariates $X_1, \dots, X_p$ (unobserved constructs), while spatial dependence is represented by an areal random effect ϕ assigned a CAR prior. Both η and ϕ enter the outcome Y either directly or indirectly through their effects on X, yielding coherent estimation of latent structure and spatial autocorrelation in one model.

Missing covariates are treated as latent variables and updated within the MCMC. At iteration t we draw

$$X_{\text{mis}}^{(t)} \sim p\big(X_{\text{mis}} \mid X_{\text{obs}},\, Y,\, \eta^{(t-1)},\, \phi^{(t-1)},\, \Theta^{(t-1)}\big)\,, \tag{9}$$

together with updates of (η, ϕ, Θ). This constitutes a Bayesian data-augmentation step in the sense of Tanner and Wong (1987); Dyk and Meng (2001): each sweep yields a completed-data draw and, by construction, uncertainty about the missing values is propagated into posterior summaries of all quantities. The strategy is consistent with modern model-based/multiple-imputation practice for complex data: sequential GLM (fully conditional specification) for mixed measurement scales (M. C. Lee and Mitra 2016), copula-based imputation for dependent series (Chapon, Ouarda, and Hamdi 2023), and MI frameworks that incorporate missing-data indicators for robustness and diagnostics (Beesley et al. 2021). Our contribution instantiates these ideas in a spatial setting by embedding the imputation mechanism inside a joint spatial factor–CAR hierarchy, rather than imputing from a separate non-spatial model.

This formulation extends random-effects factor models by placing spatial dependence directly on the latent factor (and/or area effects) instead of introducing separate group-specific loadings, thereby preserving local correlation patterns without hand-coding inter-covariate relationships. The result is particularly useful for spatial imputation, where missingness often clusters geographically and benefits from borrowing strength across neighboring areas.

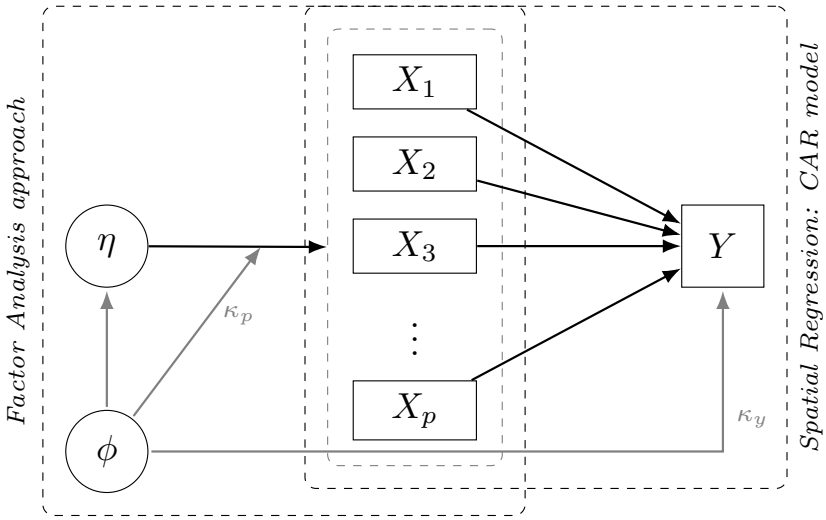

Figure 2. Propose model: integrating Bayesian missing data imputation with a spatial regression structure

Figure 2 summarises the proposed joint framework. The measurement layer uses a factor model in which a low-dimensional latent variable η captures shared variation among the covariates $X_1, \ldots, X_p$. The mapping from η to X is governed by the factor loadings λ and the measurement-error precisions, so that missing entries of X are treated as latent quantities and updated from their full conditional within the sampler. In the outcome layer, the completed covariates enter a multilevel CAR (mlvCAR) regression for $\mathbf{Y}$, where spatial dependence is modelled through an areal random effect ϕ with a CAR prior. Therefore, the response for individual j in area i is modeled as

$$y_j \mid \mathbf{X}_j, \phi_i \sim N\big(\mu_j,\, \tau_y^{-1}\big)\,, \tag{10}$$

with mean structure,

$$\mu_j = \beta_0 + \sum_{k=1}^{p} \beta_k X_{jk} + \kappa_y \phi_i. \tag{11}$$

Here, κ_y scales the contribution of the spatial effect to the outcome. This embeds spatial dependence directly within the outcome model while using the same spatial structure that governs the covariates.

Embedding spatial structure within both the latent factor and the loadings allows the model to capture spatial correlation in a principled manner. This makes the framework particularly suited for spatial imputation, where missing covariates often exhibit localised spatial patterns that traditional factor models cannot recover. Because the two layers are fit simultaneously, imputation and spatial modelling are handled as a single Bayesian hierarchy rather than as separate steps.

This integrated strategy offers several advantages over two-stage procedures. First, it propagates imputation uncertainty automatically to parameters and predictions, avoiding the incompatibilities that can arise when imputation and analysis models differ. Second, the latent factor provides dimension reduction and mitigates multicollinearity among covariates, while the mlvCAR component preserves local spatial structure by borrowing strength across neighbouring areas. Together, these features yield more stable estimates and improved predictive performance in settings with both missing data and spatial dependence.

I'm not sure I saw it, but the advantage of the kappas is also as a means to do diagnostic:
if kappa is 0 then no spatial effects for that component (X or y)

4. Study Region and Data Collection

In this study, we use Brunei-Muara as the study region. Brunei-Muara, the most urbanised district in Brunei Darussalam, consists of 18 mukim (sub-districts), but for this study, only 12 are considered. The exclusion of 6 mukim is due to their location in watery areas where house sales do not occur.

The dataset contains $N = 2284$ property transactions recorded since 2015 (Jamil et al. (2025)), specifically focusing on landed houses with a plot area of less than 0.24 hectares.

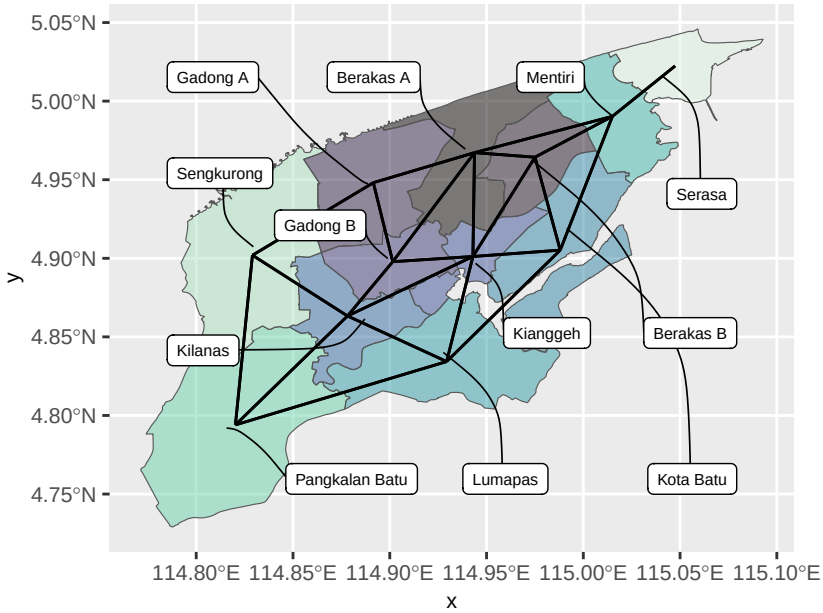

Figure 3. Brunei-Muara map and its neighbourhood structure

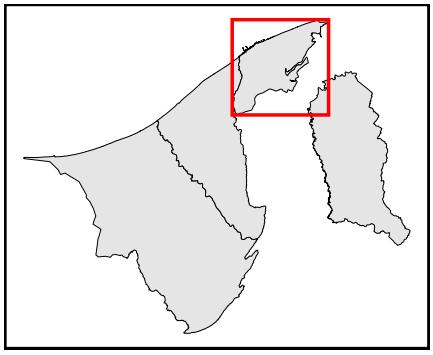

Figure 4. Brunei-Muara in Brunei Region

Figure 5. Brunei in the South-east Asia region

Table 1. Summary of Brunei-Muara House Price Dataset: Observations, Missing Values, and Mean Values per Mukim. It consists of $n = 2284$ observations with total of 895 missing value in covariates

Mukim	N	Plot area	Floor area	Beds	Baths
Berakas A	23	0.113 (0.040) – 6	2401 (503) – 1	4.43 (0.75) – 2	4.43 (0.98) – 2
Berakas B	198	0.095 (0.035) – 8	2766 (497) – 28	4.31 (0.58) – 15	4.19 (1.07) – 15
Gadong A	46	0.100 (0.042) – 3	2994 (738) – 7	4.07 (0.80) – 0	4.39 (1.18) – 0
Gadong B	153	0.083 (0.040) – 19	2564 (720) – 23	4.32 (0.62) – 6	4.12 (1.20) – 6
Kianggeh	21	0.163 (0.069) – 10	4024 (1488) – 9	4.62 (1.26) – 5	4.88 (2.16) – 5
Kilanas	349	0.097 (0.036) – 37	2395 (578) – 68	4.05 (0.59) – 20	3.59 (0.97) – 19
Kota Batu	143	0.090 (0.029) – 23	2556 (640) – 45	4.12 (0.59) – 5	3.69 (0.96) – 5
Lumapas	245	0.102 (0.036) – 23	1855 (356) – 32	3.74 (0.56) – 9	3.07 (0.76) – 10
Mentiri	224	0.098 (0.035) – 21	2209 (419) – 34	4.12 (0.67) – 8	3.90 (0.94) – 6
Pangkalan Batu	207	0.104 (0.032) – 15	2052 (451) – 71	3.86 (0.50) – 5	3.14 (0.76) – 4
Sengkurong	496	0.114 (0.044) – 75	2305 (687) – 84	4.07 (0.64) – 16	3.49 (1.03) – 16
Serasa	179	0.098 (0.048) – 24	2427 (801) – 34	4.25 (0.61) – 8	4.01 (1.39) – 8

Note:
Each cell shows the mean and standard deviation in parentheses, followed by the number of missing values for that covariate, within each mukim.

Table 1 presents the descriptive summary of key housing attributes across 12 *mukim*, namely plot area, floor area, number of bedrooms, and number of bathrooms. All variables are summarised using the median and interquartile range ($Q_1 - Q_3$) to capture the central tendency and variability, given the likely skewed distribution of housing characteristics.

The plot area exhibits notable spatial variation. Urban mukim such as Gadong A and Gadong B report the smallest median plot sizes (0.09 and 0.07, respectively), whereas Kianggeh records the largest median plot area (0.22), suggesting a spatial gradient in land availability or land use intensity. Similar patterns are observed in floor area, with Kianggeh again having the highest median floor size (4,218), in contrast to Lumapas (1,834) and Pangkalan Batu (2,000), which report comparatively smaller built-up areas. These findings may reflect differences in housing typologies, density, or urban development patterns across mukim.

The number of bedrooms is relatively consistent across the sub-districts, with a median of 4.00 in most locations. Kianggeh reports a slightly higher median (4.5), with an upper quartile of 6.00, indicating the presence of larger dwelling units in that area. The number of bathrooms also demonstrates some variability. While the majority of mukim report medians between 3 and 4 bathrooms, Kianggeh again distinguishes itself with a higher median of 5.5 and an interquartile range extending to 7.0, further suggesting more spacious or higher-end housing stock.

In terms of missing data, the highest proportions are observed in the floor area and plot area variables, particularly in Sengkurong (84 missing values), Kilanas (68), and Gadong B (45). Bedrooms and bathrooms generally have fewer missing entries, although some mukim such as Berakas B and Pangkalan Batu still report moderate levels of missingness. These patterns highlight potential limitations in data completeness, which may need to be addressed in subsequent analyses through our propose imputation method.

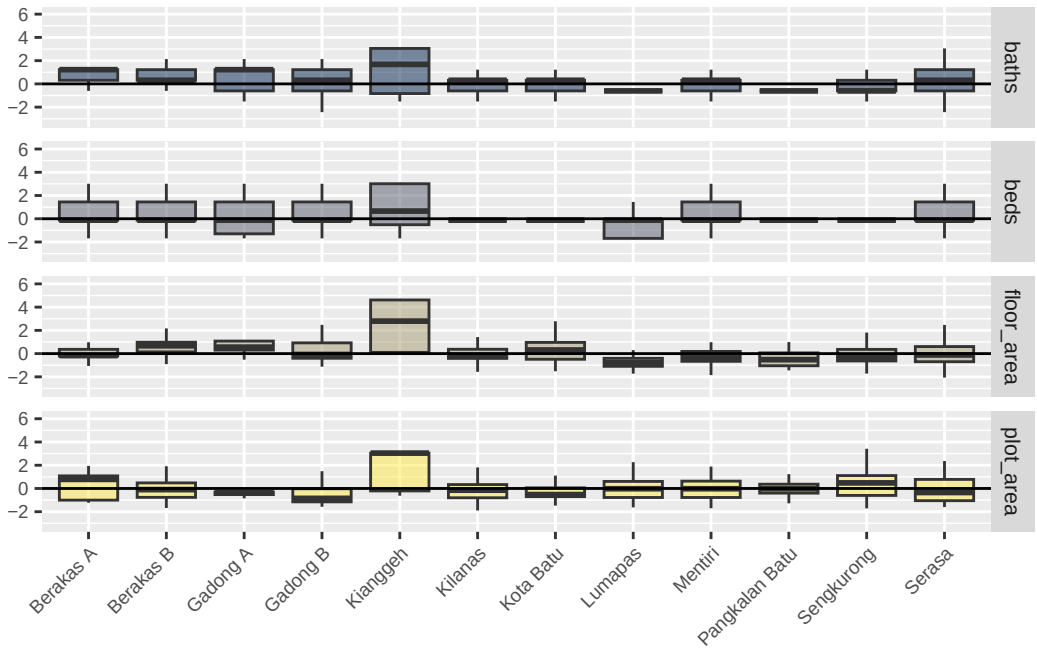

Figure 6. Distribution of covariates by Mukim

Figure 6 complements the tabular summary by presenting the standardised distributions of key housing covariates—plot area, floor area, number of bedrooms, and number of bathrooms—across the 12 mukim. The plot clearly illustrates the spatial heterogeneity of housing characteristics. For instance, Kianggeh stands out across all panels, with notably higher standardized values, particularly in floor_area and plot_area, reinforcing its distinct profile identified in Table 1. This suggests that residential units in Kianggeh tend to be substantially larger in both land and built-up dimensions compared to other mukim.

Other mukim such as Gadong B and Sengkurong exhibit moderate positive deviations in some variables, such as a slightly higher number of bathrooms, while Lumapas, Pangkalan Batu, and Kota Batu consistently show lower or median-centered distributions, indicating more compact and uniform housing structures.

Moreover, the relatively narrow spread and symmetric boxplots for variables like beds and baths across most mukim indicate a consistent distribution pattern, with few extreme values. In contrast, plot area and floor area show greater variability and more pronounced outliers in certain locations, hinting at a wider range of property types and lot sizes.

Overall, the figure visually affirms the patterns noted in the summary statistics and highlights the variation in housing characteristics that may be attributable to differing urban development intensities, socio-economic profiles, or land-use regulations across mukim.

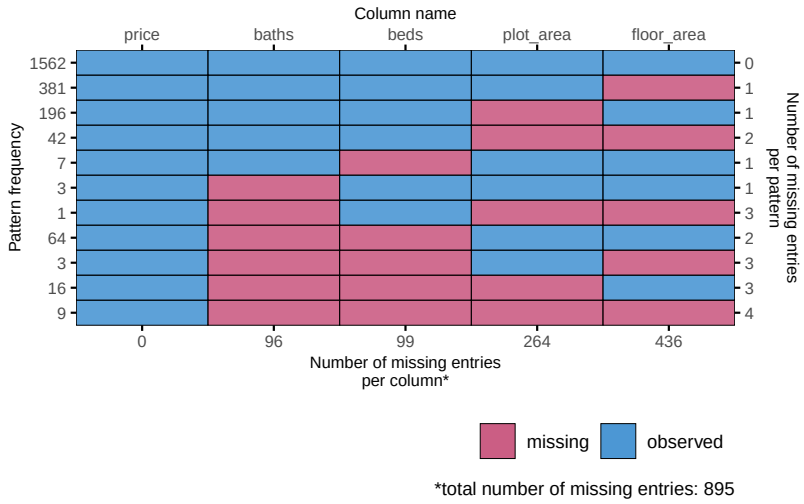

Figure 7. Missing values pattern from original house prices data in Brunei-Muara

The observed pattern as in Figure 7 revealed that missing values tend to occur more frequently in certain variables, often following a systematic structure. Notably, whenever `baths` is missing, `beds` is also missing, indicating a systematic missingness structure. This type of dependency is characteristic of the MAR mechanism, where the probability of missingness is related to other observed variables rather than being completely random.

Overall, the summary reveals both intra- and inter-mukim heterogeneity in property characteristics, with Mukim Kianggeh consistently standing out for its larger and more amenity-rich housing stock. These patterns underscore the importance of accounting for spatial variability when modeling residential property characteristics in the Brunei-Muara region.

5. Experimental design and method

As stated in the previous section, the Brunei-Muara house price dataset contains nearly 40% missing values in its covariates, highlighting the need for a robust imputation strategy. To address this issue, study designs two experiments aimed at evaluating the effectiveness of spatial factor analytic models in imputing missing values within the covariates of house price datasets.

The first experiment aims to assess the suitability and necessity of incorporating spatial factor analysis for such tasks. We begin by generating spatial datasets with artificially induced missing values in the covariates, based on characteristics derived from a preliminary analysis of the Brunei-Muara dataset. Two missing data mechanisms are considered: Missing Completely at Random (MCAR) and Missing at Random (MAR), each applied at four levels of missingness (10%, 20%, 40%, and 80%). Three models are fitted across each scenario: (a) a spatial factor analysis model with area-specific latent variables (see Equation 7), (b) a simplified spatial factor model with a common measurement structure (in this case, $\kappa = 0$), and (c) a standard non-spatial factor analysis model (see **?@eq-factmod3**). For each scenario, the simulation is repeated 100 times,

and the results are summarised in terms of average posterior mean or mode estimates. Model performance is evaluated using standard metrics such as bias, root mean squared error (RMSE), and coverage of the recovered parameters.

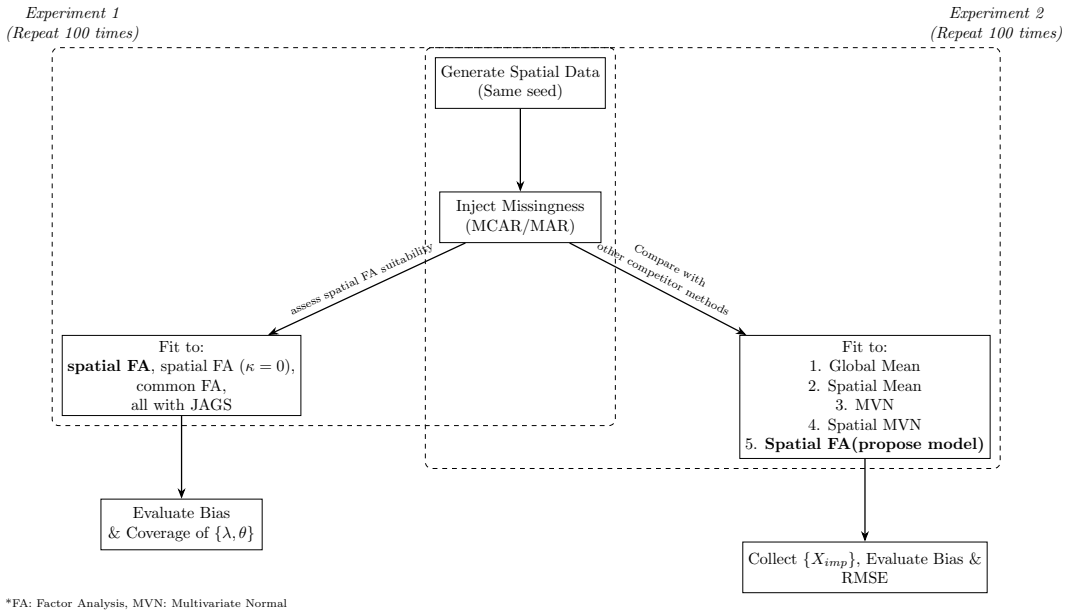

Figure 8. Overview of simulation experiments with shared seed in data generation and missingness injection

The second experiment (Figure 8) is designed to benchmark the proposed spatial factor analysis model against several widely used imputation methods. Using the same simulated datasets as in the first experiment—each incorporating MCAR and MAR mechanisms with varying degrees of missingness—we compare five approaches: global mean imputation (Paik 1997), spatially stratified mean imputation (e.g., Jamil 2025), multivariate normal (MVN) imputation based on standard procedures K. J. Lee and Carlin (2010), MVN imputation enhanced with spatial dependence Lin et al. (2024), and the proposed spatial factor analysis model. Each method is assessed through 100 repetitions per scenario. The primary evaluation metric is bias, with additional assessment using root mean squared error (RMSE) to capture overall imputation accuracy.

Following the simulation study, the methodology is then applied to real-world house price data from Brunei-Muara, a district in Brunei Darussalam. The dataset includes property sales records along with covariates such as plot area, floor area, number of bedrooms, and number of bathrooms. The analysis consists of two primary steps. First, the mlvCAR model is fitted separately to both the complete Brunei-Muara dataset and its incomplete counterpart, where missing values are introduced following a MAR pattern informed by the original dataset. This step allow us to assess the impact of missing data on model estimation. Second, performance evaluation is conducted by comparing key aspects: (1) the bias in model parameters, including β, ρ, τ_ϕ, and τ_y ; (2) the bias in area-level covariate estimates to understand how missing data and imputation influence spatial unit estimates; and (3) the predicted house prices to measure the effect of missing data on price estimations.

Through this combined simulation and empirical application, we aims to validate the

robustness of the factor analysis-based imputation method before applying it to real-world spatial property data.

6. Results

Results from both the simulation study and the real-world case study on house prices in Brunei-Muara are presented. It is important to note that this study involves two similarly abbreviated terms, mlvCAR and MCAR which refer to entirely different concepts. While both have been explained in previous sections, we emphasize here that mlvCAR denotes the multilevel Conditionally Autoregressive model used for spatial regression, whereas MCAR refers to Missing Completely at Random, a missing data mechanism.

6.1. *Experiment on spatial data imputation based on a factor analysis model*

Table 4 and Table 5 present the bias estimates for the λ_p and error variances, θ_p under different missing data setting, MCAR and MAR, with missingness proportions ranging from 10% to 80%. In this context, Model Spatial FA refers to a spatial factor analysis, refer to Equation 6, second one is a spatial factor analysis with $\kappa = 0$ (i.e., no spatial smoothing), and FA is a standard factor analysis with no spatial structure (Equation 4).

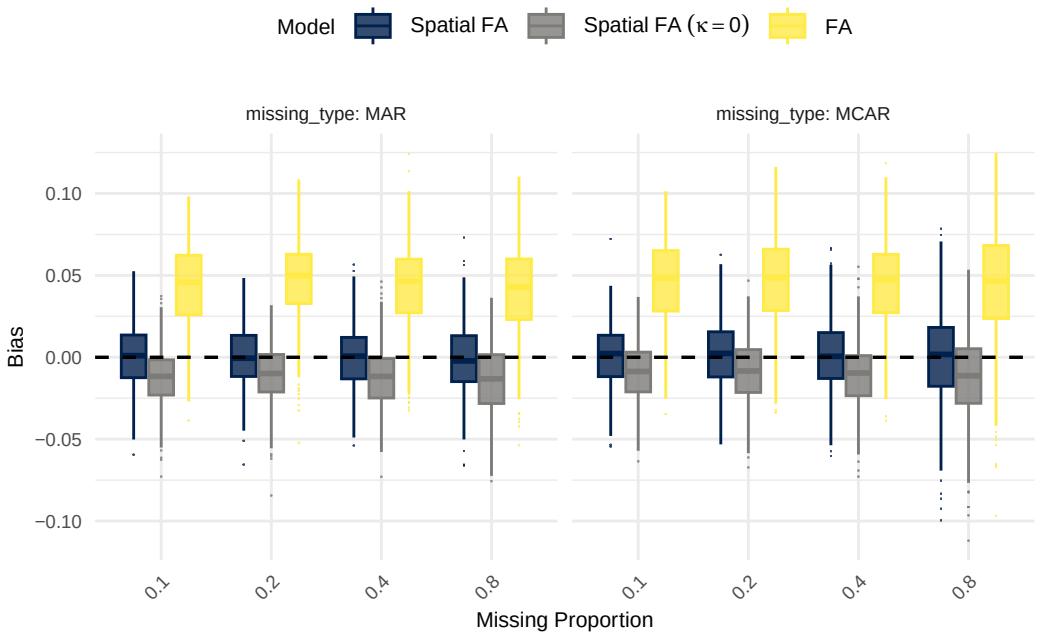

Figure 9. Distribution of factor loading (λ) bias across replications

Figure 9 shows clear separation among the three approaches. Spatial FA has medians essentially at zero across both MAR and MCAR and for all missingness levels (0.1–0.8); its interquartile ranges are the narrowest, with few extreme values. Spatial FA ($\kappa = 0$)

exhibits a small but systematic negative bias that becomes more pronounced as the missing proportion increases; dispersion is modest at low missingness but widens at 0.8. The non-spatial FA tends to have a positive bias and the largest spread, with variability and outliers increasing with the amount of missing data.

Overall, these patterns indicate that modeling spatial structure in the factor layer (with $\kappa > 0$) yields the most stable and nearly unbiased estimates of λ. Constraining $\kappa = 0$ induces mild underestimation, and omitting spatial information altogether leads to greater variability and overestimation.

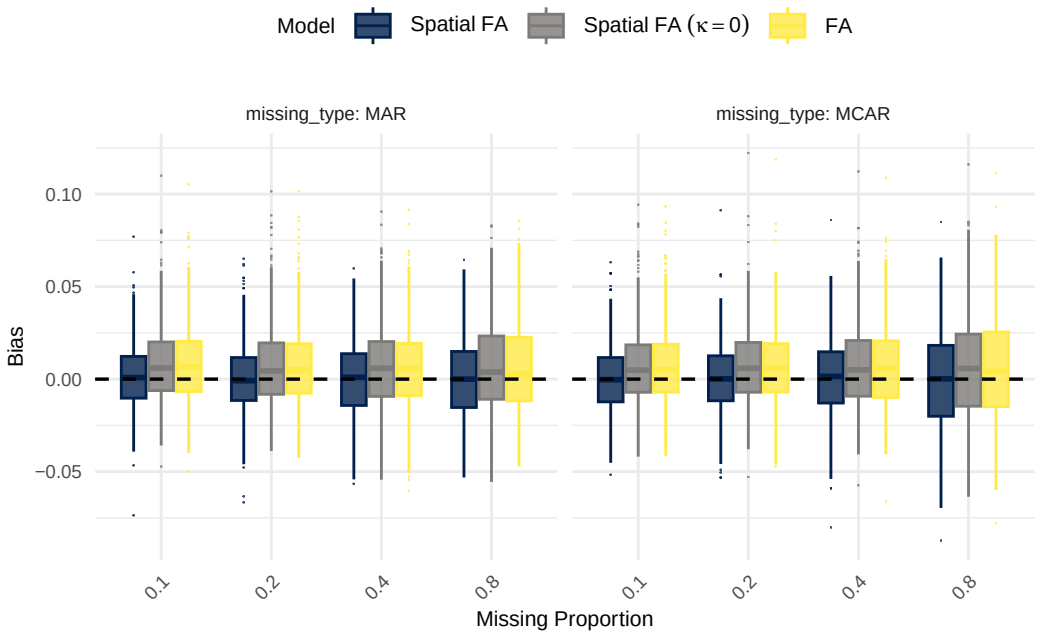

Figure 10. Distribution of error variance (θ) bias across replications

A similar pattern holds for θ_p, as shown in Table 5 and visualised in Figure 10. Spatial FA maintains near-zero median bias with the narrowest 95% posterior intervals across both MAR and MCAR and for all missingness levels. Spatial FA ($\kappa = 0$) shows a small positive bias that grows as the proportion of missingness increases, with intervals widening at higher levels. The non-spatial FA consistently exhibits the largest bias for θ_p, especially under MCAR, together with the widest intervals and more outliers.

These results underscore the value of modelling spatial structure in the factor layer: the full Spatial FA yields the most accurate and stable estimates, whereas removing spatial coupling (non-spatial FA) or constraining it ($\kappa = 0$) degrades performance as missingness increases.

6.2. *Comparing the spatial factor analysis method with other competitor method*

Figure 11 shows the distribution of bias in the imputed covariate values (X_m) across different methods and missing data scenarios. Each subplot represents a different imputation method: Spatial Factor Analysis, Global Mean, Spatial Mean, Multivariate

Normal (MVN), and Spatial MVN. Within each panel, boxplots display the bias under varying missing scenarios and proportions.

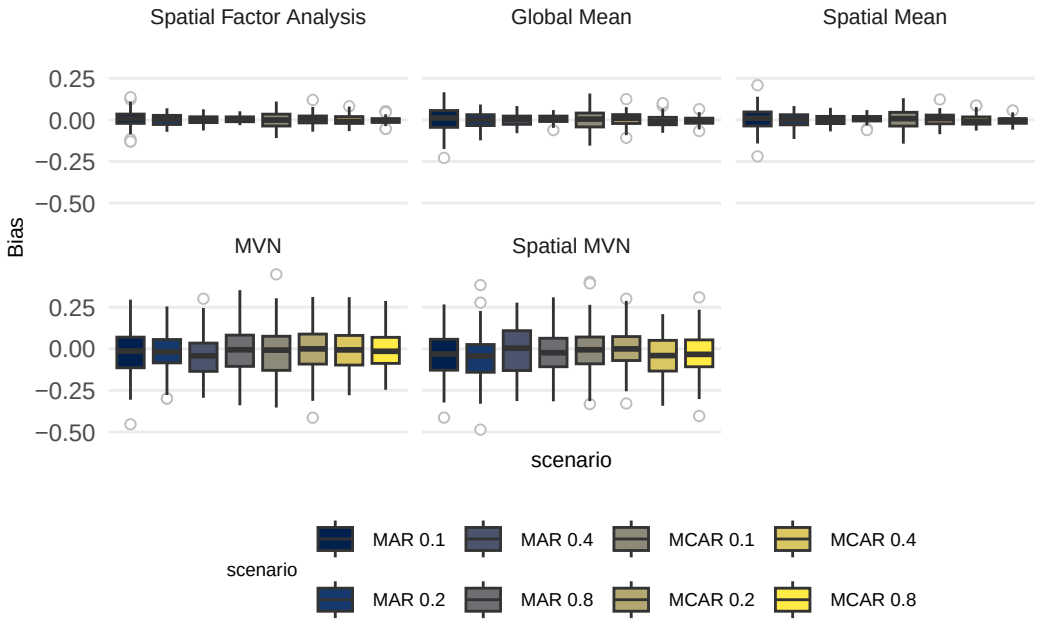

Figure 11. Distribution of X Bias Across Replications

The Spatial Factor Analysis method exhibits consistently low and stable bias across all simulation scenarios, indicating robust performance in the presence of missing data. Although the Global Mean and Spatial Mean approaches also demonstrate relatively low bias, their distributions appear moderately more dispersed. This suggests that while averaging-based imputations offer a degree of stability, they may be limited in capturing nuanced spatial dependencies. In contrast, the MVN and Spatial MVN methods tend to show greater variability, especially under MAR mechanisms with substantial data loss. These findings highlight important distinctions in how each method incorporates structural assumptions to address missing covariate information within spatial modeling contexts.

6.3. *Case study: Brunei-Muara house prices data*

We applied the proposed methodological framework (see Figure 2) to the Brunei-Muara house price dataset to evaluate its performance under structured missing data conditions. To begin, we derived a complete dataset by removing all rows containing missing values from the original dataset. We then induced missingness into the complete dataset based on the empirical missing data pattern observed in the original data, resulting in a second dataset, let say: incomplete dataset that closely mirrors the real-world missingness structure (see Figure 7 and Figure 12).

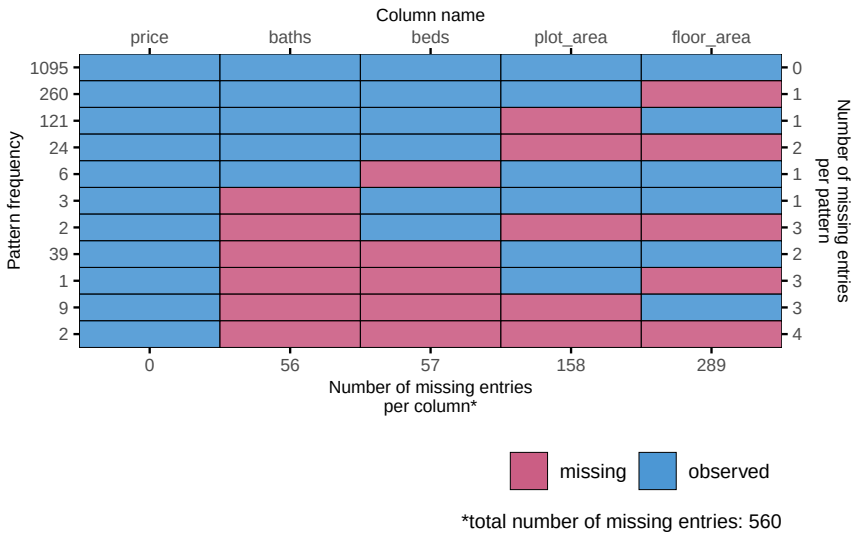

Figure 12. Induced missing values: adapted pattern from the original data. This plot illustrates the missing values across key variables (beds, baths, plot area and floor area)

The Brunei-Muara incomplete dataset used in this analysis comprises $n = 1562$ observations, with a missingness proportion exceeding 35%. As shown in Table 2, this dataset was carefully constructed to closely resemble the original complete dataset described in Table 1, ensuring consistency in distributional patterns and structural characteristics.

Table 2. Summary of Brunei-Muara House Price Dataset that is induced with missing value (incomplete dataset). It contain $n = 1562$ observations with a total of 560 missing entries across covariates

Mukim	N	Plot area	Floor area	Beds	Baths
Berakas A	15	0.121 (0.038) – 1	2554 (475) – 3	4.20 (0.77) – 0	4.33 (1.05) – 0
Berakas B	150	0.092 (0.033) – 12	2737 (434) – 33	4.28 (0.57) – 4	4.19 (1.15) – 5
Gadong A	37	0.102 (0.045) – 2	3006 (710) – 5	4.27 (0.73) – 0	4.75 (1.02) – 1
Gadong B	109	0.084 (0.042) – 14	2556 (737) – 19	4.31 (0.63) – 6	4.09 (1.32) – 6
Kianggeh	11	0.150 (0.070) – 2	4287 (1492) – 1	4.60 (1.51) – 1	4.60 (2.55) – 1
Kilanas	233	0.098 (0.039) – 25	2395 (555) – 43	4.05 (0.57) – 11	3.60 (0.98) – 11
Kota Batu	74	0.091 (0.028) – 5	2755 (548) – 18	4.25 (0.52) – 2	3.72 (1.17) – 2
Lumapas	188	0.096 (0.031) – 17	1839 (359) – 37	3.73 (0.52) – 6	2.98 (0.61) – 6
Mentiri	168	0.098 (0.038) – 22	2167 (366) – 29	4.06 (0.65) – 6	3.81 (0.93) – 6
Pangkalan Batu	121	0.103 (0.021) – 17	2061 (464) – 21	3.92 (0.40) – 3	3.07 (0.62) – 2
Sengkurong	336	0.113 (0.046) – 30	2313 (696) – 64	4.09 (0.60) – 10	3.44 (1.00) – 9
Serasa	120	0.098 (0.051) – 11	2443 (809) – 16	4.36 (0.63) – 8	4.19 (1.55) – 7

Note:
Each cell shows the mean and standard deviation in parentheses, followed by the number of missing values for that covariate, within each mukim.

A simple spatial regression model is specified to capture the relationship between house prices and key property characteristics

$$\texttt{price} = \beta_0 + \beta_1\texttt{plot_area} + \beta_2\texttt{floor_area} + \beta_3\texttt{beds} + \beta_4\texttt{baths} \tag{12}$$

This formulation reflects a standard hedonic pricing approach, extended within a spatial

Table 3. Bias summary and RMSE for model parameters

	Parameter	**Bias [95% CI]**	**RMSE**
Regression coefficients	β_1	0.016 [-0.009, 0.042]	0.021
	β_2	0.011 [-0.024, 0.044]	0.021
	β_3	-0.014 [-0.046, 0.015]	0.022
	β_4	-0.010 [-0.041, 0.021]	0.019
Factor loadings	λ_1	-0.011 [-0.074, 0.054]	0.034
	λ_2	-0.002 [-0.057, 0.054]	0.028
	λ_3	0.008 [-0.039, 0.057]	0.026
	λ_4	0.013 [-0.039, 0.070]	0.031
Variance of the independent noise	θ_1	-0.000 [-0.057, 0.063]	0.031
	θ_2	-0.005 [-0.044, 0.038]	0.022
	θ_3	-0.012 [-0.048, 0.028]	0.023
	θ_4	-0.011 [-0.049, 0.030]	0.023
Scaling factor to spatial influence strength	κ_1	0.005 [-0.155, 0.180]	0.085
	κ_2	-0.005 [-0.130, 0.119]	0.064
	κ_3	0.002 [-0.128, 0.143]	0.068
	κ_4	0.030 [-0.113, 0.172]	0.080
Spatial autocorrelation	ρ	-0.052 [-0.234, 0.467]	0.196
Variance Components	τ^2	-0.027 [-0.338, 0.413]	0.213
	ν^2	-0.002 [-0.013, 0.011]	0.006

framework to account for the influence of land size, built-up area, and interior features on property prices.

We applied our integrated modelling framework, which combines spatial factor analysis-based imputation with a multilevel Conditional Autoregressive (mlvCAR) model, to both the complete and incomplete dataset. The framework was implemented using the JAGS environment, allowing simultaneous estimation of missing covariate values, regression parameters, and spatial random effects. The analysis using the complete data serves as a reference, representing inference under fully observed covariates, while the results from incomplete dataset demonstrate the model's performance when handling structured missingness.

Table 3 presents the summary of posterior bias and root mean square error (RMSE) for the model parameters when using the incomplete Brunei-Muara dataset.

Across β_p, bias values are relatively small, with median biases close to zero and narrow 95% credible intervals, indicating stability in estimation despite missing data. The λ_p and θ_p show similarly modest biases and low RMSEs, suggesting that the factor structure is well recovered post-imputation.

For parameters that govern the spatial structure, such as the scaling factor κ_p and ρ, the results remain reasonably accurate, though slight increases in RMSE (e.g., 0.196 for rho) are observed. The variance components, particularly τ^2, exhibit the highest RMSE (0.213), indicating more uncertainty in estimating spatially structured random effects under incomplete data conditions. This observation aligns with well-established findings in the literature that variance components are inherently more difficult to estimate accurately, particularly in hierarchical Bayesian models (Gelman (2004)). This phenomenon can be attributed to the additional uncertainty introduced during the imputation phase. While the factor analysis-based imputation effectively recovers missing covariates, it also injects variability into the data, which propagates into the estimation of variance parameters and hinders their precision. In this context, the extra variability

from the imputation contributes to the increased bias observed in τ^2 and ν^2, underscoring the complexity of accurately estimating variance components in the presence of missing data and spatial structure.

In general, the combination of factor-analytic imputation and CAR modelling demonstrates strong robustness in estimating most parameters, particularly those essential for interpreting fixed effects and spatial dependence.

Figure 14 is presented to illustrate the comparison, with credible intervals from both datasets overlaid. A considerable degree of overlap is observed, indicating that the underlying structure of the data was effectively recovered through the imputation process.

In summary, robust parameter estimation and model fit are maintained even under substantial missingness. These findings suggest that meaningful spatial analysis can still be achieved when missing data are handled appropriately.

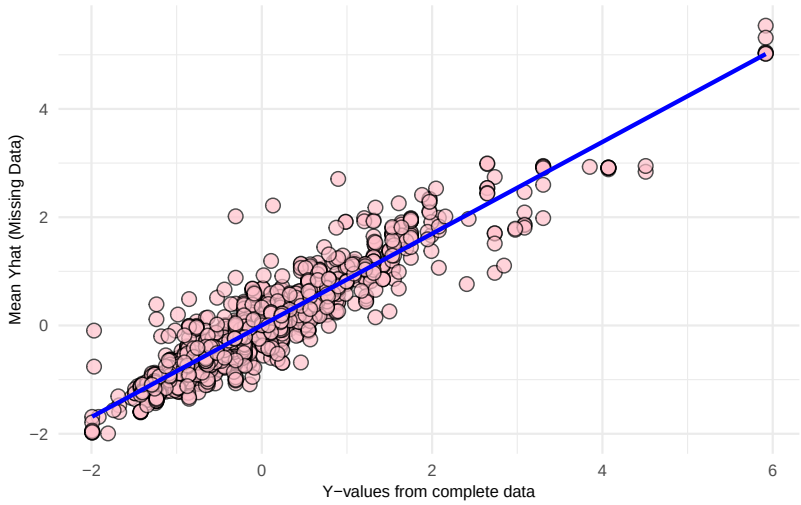

Figure 13. Scatter plot Mean Y Estimate from missing data vs. Y-values from complete data

Furthermore, the scatter plot in Figure 13 help us to gain insight about comparison between y from complete data (true values) and the mean posterior predictions, $\hat{y}$ obtained from the dataset with missing covariates after applying the factor analysis-based imputation method. The strong alignment along the 45-degree line (blue) indicates that the predictions from the imputed dataset closely match those from the complete dataset, suggesting that the imputation approach preserves the underlying structure of the data. The RMSE of 0.148 further also confirms the small deviation between the two estimates, demonstrating that the imputation process successfully mitigates missing data effects without introducing significant bias.

Figure 14 presents caterpillar plots of the posterior bias estimates across all imputed values, stratified by *Mukim*. Each point represents the posterior mean bias, with horizontal error bars denoting 95% credible intervals. A vertical red dashed line at zero aids visual comparison.

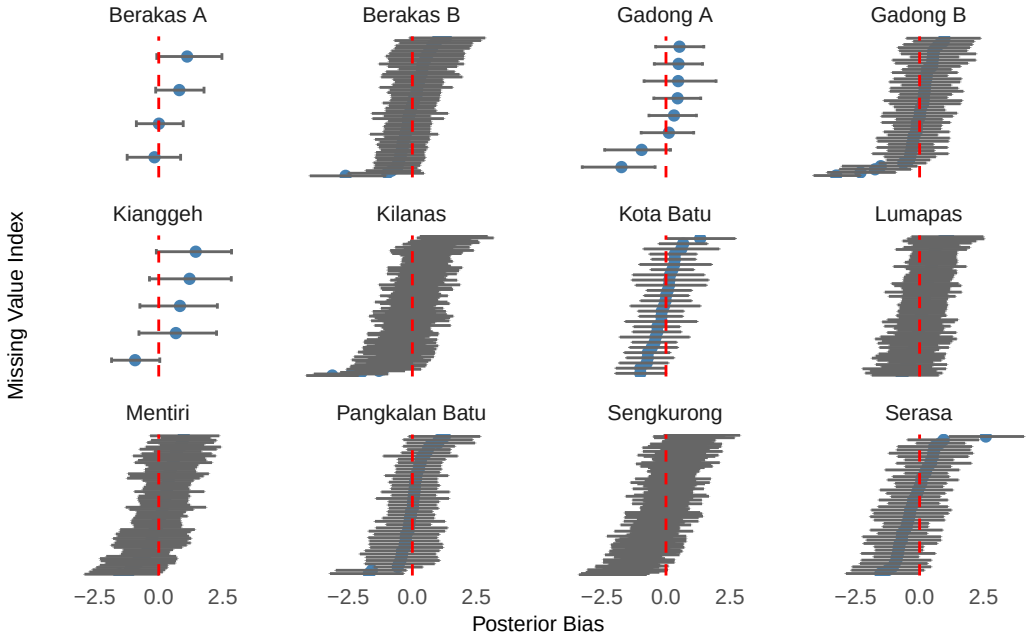

Figure 14. Posterior bias estimates with 95% credible intervals for each imputed value, grouped by mukim. Each point denotes the posterior mean bias, and error bars represent the corresponding 95% credible interval. The dashed red line at zero indicates unbiased estimation

In most mukim, the biases are symmetrically distributed around zero, and the credible intervals generally contain the true value, suggesting effective recovery of missing values. Notably, no substantial systematic over or under-estimation is observed across spatial regions, reinforcing the robustness of the imputation and model performance under varying missingness patterns.

7. Discussion

This study demonstrates the effectiveness of integrating factor analysis-based imputation with a multilevel Conditional Autoregressive (mlvCAR) model using Leroux priors to address missing covariate information in spatially hierarchical house price data. Through a series of simulation studies and a real-data application, the analysis highlights how different imputation strategies impact both parameter recovery and predictive performance when faced with varying degrees and mechanisms of missingness.

In the first experiment, we compared three variants of factor analysis-based imputation: the proposed Spatial Factor Analysis (Spatial FA), a variant without spatial smoothing ($\kappa = 0$), and traditional non-spatial Factor Analysis (FA). The proposed Spatial FA consistently produced lower and more stable bias in estimating key parameters, particularly factor loadings (λ) and specific variances (θ). The inclusion of spatial structure allowed the model to borrow information across neighboring units, which enhanced robustness in sparse settings. Removing spatial smoothing (as in the $\kappa = 0$ case) reduced this benefit, while traditional FA showed the highest bias and variability.

The second experiment compared Spatial FA against four other commonly used im-

putation approaches: Global Mean, Spatial Mean, Multivariate Normal (MVN), and Spatial MVN imputation. While Global and Spatial Mean imputations performed reasonably well, their bias distributions were slightly more dispersed, suggesting limitations in adapting to spatial heterogeneity, especially under Missing at Random (MAR) mechanisms. As the proportion of missingness increased, MVN and Spatial MVN methods exhibited even greater variability in bias. These patterns highlight the importance of spatial structure in imputation, especially in datasets with complex underlying geography and multilevel nesting.

An unexpected but insightful finding emerged when examining the behavior of the imputed covariate values (X_m). The dispersion in bias for X_m appeared to decrease as the proportion of missingness increased. While counterintuitive, this pattern likely reflects the model's increasing reliance on structural assumptions, such as spatial smoothing and low-rank latent structure, when observed data are sparse. With fewer data points to inform estimation, the model "fills in" gaps using smoother or more averaged values, which reduces variability but does not necessarily improve accuracy. This is akin to painting a landscape with only vague outlines, the resulting picture may look consistent, but lacks fidelity to the underlying variation. In contrast, the bias in estimated parameters like loadings (λ) and θ increased with higher missingness, as expected, since parameter estimation depends on having sufficient variation in the covariates.

These results support the view that integrating imputation within the modelling framework, mainly one that is spatially aware, yields more coherent and context-sensitive inference. While Baker, White, and Mengersen (2014) found that simple mean imputation performed surprisingly well in their spatial health data (particularly under structured missingness), they also noted that the effectiveness of imputation is highly dependent on the spatial and contextual structure of the data. Our findings support and extend this insight by demonstrating that a modular framework, combining factor analysis with spatial smoothing, can achieve comparable or superior performance without explicitly modeling the missingness mechanism.

The third experiment, involving real property data from Brunei-Muara, further demonstrated the practicality and adaptability of the proposed method. When applied to datasets with missing covariate values, the integrated Spatial FA imputation and mlv-CAR modelling in JAGS yielded regression coefficients and predicted prices that closely aligned with those derived from the complete dataset, as indicated by a low RMSE. The preservation of spatial structure in the imputed covariates is evidenced by the consistent patterns observed in the caterpillar plots of posterior bias across mukim (Figure 13). These plots reveal that the imputation model captured regional variation in bias, with several mukim exhibiting coherent deviations from zero. This spatial heterogeneity in imputation accuracy suggests that the model retained local contextual information during the imputation process, reinforcing the claim that spatial structure was effectively preserved.

However, the variance parameter τ^2 exhibited relatively higher mean bias compared to the regression coefficients. This finding aligns with the well-known difficulty of estimating variance components in hierarchical models, as discussed by Gelman (2004). In this context, the additional uncertainty introduced during the imputation phase may further compound the challenge. Although factor analysis-based imputation effectively preserves underlying covariate relationships, it introduces an extra layer of variability that propagates into the estimation of precision parameters. Nonetheless, the overall model performance remained robust, with clear evidence that the proposed framework

adapts well to spatial heterogeneity while handling missingness in a principled manner.

The findings suggest that the proposed approach offers a computationally efficient, spatially coherent, and robust strategy for handling missing covariates in hierarchical spatial models. The factor analysis component reduces the dimensionality of the covariate space, making the approach scalable, while the CAR structure ensures spatial continuity is maintained. This balance between parsimony and spatial sensitivity makes it particularly well-suited to real-world applications with limited data and strong geographic context.

Future work could explore extensions such as dynamic imputation for spatio-temporal data, incorporation of alternative spatial priors, or robustness analyses under non-ignorable missingness mechanisms. Overall, the findings contribute to the growing methodological toolkit for addressing missing data in spatial regression and demonstrate the potential of spatially-structured factor models for improving inference in complex, multilevel spatial settings.

Appendix

Table 4. Bias Estimates for lambda. Mean values with 95% confidence intervals

Missing Type	Missing (%)	Mean Bias	95% CI
Spatial FA			
MAR	0.1	0.00005	[-0.038, 0.039]
MAR	0.2	0.00043	[-0.038, 0.039]
MAR	0.4	0.00030	[-0.039, 0.041]
MAR	0.8	-0.00113	[-0.045, 0.044]
MCAR	0.1	0.00064	[-0.037, 0.039]
MCAR	0.2	0.00188	[-0.037, 0.042]
MCAR	0.4	0.00028	[-0.041, 0.043]
MCAR	0.8	0.00070	[-0.049, 0.052]
Spatial FA($\kappa = 0$)			
MAR	0.1	-0.01240	[-0.047, 0.022]
MAR	0.2	-0.01030	[-0.045, 0.025]
MAR	0.4	-0.01259	[-0.049, 0.024]
MAR	0.8	-0.01332	[-0.054, 0.028]
MCAR	0.1	-0.00988	[-0.045, 0.025]
MCAR	0.2	-0.00927	[-0.045, 0.027]
MCAR	0.4	-0.01133	[-0.05, 0.028]
MCAR	0.8	-0.01206	[-0.059, 0.035]
FA			
MAR	0.1	0.04322	[0.006, 0.081]
MAR	0.2	0.04570	[0.008, 0.084]
MAR	0.4	0.04222	[0.003, 0.082]
MAR	0.8	0.04074	[-0.004, 0.086]
MCAR	0.1	0.04576	[0.008, 0.084]
MCAR	0.2	0.04643	[0.008, 0.086]
MCAR	0.4	0.04450	[0.003, 0.087]
MCAR	0.8	0.04361	[-0.007, 0.095]

Table 5. Bias Estimates for theta. Mean values with 95% confidence intervals

Missing Type	Missing (%)	Mean Bias	95% CI
Spatial FA			
MAR	0.1	0.00172	[-0.032, 0.037]
MAR	0.2	0.00032	[-0.035, 0.037]
MAR	0.4	0.00004	[-0.037, 0.039]
MAR	0.8	0.00005	[-0.043, 0.045]
MCAR	0.1	0.00045	[-0.034, 0.036]
MCAR	0.2	0.00038	[-0.035, 0.038]
MCAR	0.4	0.00038	[-0.038, 0.041]
MCAR	0.8	-0.00029	[-0.05, 0.051]
Spatial FA($\kappa = 0$)			
MAR	0.1	0.00841	[-0.026, 0.045]
MAR	0.2	0.00727	[-0.028, 0.044]
MAR	0.4	0.00671	[-0.031, 0.046]
MAR	0.8	0.00701	[-0.037, 0.053]
MCAR	0.1	0.00704	[-0.028, 0.044]
MCAR	0.2	0.00746	[-0.029, 0.045]
MCAR	0.4	0.00722	[-0.032, 0.049]
MCAR	0.8	0.00685	[-0.044, 0.06]

(continued)

Missing Type	Missing (%)	Mean Bias	95% CI
FA			
MAR	0.1	0.00821	[-0.027, 0.045]
MAR	0.2	0.00712	[-0.029, 0.045]
MAR	0.4	0.00654	[-0.032, 0.046]
MAR	0.8	0.00681	[-0.038, 0.054]
MCAR	0.1	0.00685	[-0.029, 0.044]
MCAR	0.2	0.00721	[-0.03, 0.046]
MCAR	0.4	0.00696	[-0.033, 0.049]
MCAR	0.8	0.00633	[-0.046, 0.061]

Table 6. Summary of Bias Across Imputation Methods and Missingness Scenarios

Scenario	Min	Q1	Median	Q3	Max	Mean
Spatial FA						
MAR 0.1	-0.1304	-0.0208	0.0081	0.0352	0.1364	0.0048
MAR 0.2	-0.0715	-0.0275	0.0047	0.0273	0.0699	0.0001
MAR 0.4	-0.0637	-0.0161	0.0031	0.0187	0.0636	0.0017
MAR 0.8	-0.0317	-0.0109	0.0046	0.0182	0.0520	0.0042
MCAR 0.1	-0.1090	-0.0368	-0.0008	0.0347	0.1104	0.0009
MCAR 0.2	-0.0707	-0.0180	0.0053	0.0236	0.1204	0.0037
MCAR 0.4	-0.0672	-0.0209	-0.0086	0.0201	0.0820	-0.0027
MCAR 0.8	-0.0541	-0.0149	-0.0047	0.0079	0.0525	-0.0030
Global Mean						
MAR 0.1	-0.2284	-0.0448	0.0111	0.0565	0.1661	0.0053
MAR 0.2	-0.1228	-0.0341	-0.0001	0.0315	0.0924	-0.0011
MAR 0.4	-0.0799	-0.0267	0.0038	0.0231	0.0834	-0.0003
MAR 0.8	-0.0614	-0.0096	0.0070	0.0230	0.0590	0.0065
MCAR 0.1	-0.1548	-0.0423	0.0048	0.0414	0.1584	0.0018
MCAR 0.2	-0.1073	-0.0215	0.0098	0.0319	0.1247	0.0046
MCAR 0.4	-0.0776	-0.0291	-0.0083	0.0155	0.1006	-0.0044
MCAR 0.8	-0.0666	-0.0200	-0.0044	0.0105	0.0647	-0.0042
Spatial Mean						
MAR 0.1	-0.2189	-0.0366	0.0103	0.0480	0.2083	0.0057
MAR 0.2	-0.1146	-0.0297	0.0028	0.0313	0.0833	0.0004
MAR 0.4	-0.0691	-0.0219	0.0037	0.0214	0.0727	0.0012
MAR 0.8	-0.0612	-0.0064	0.0087	0.0206	0.0593	0.0059
MCAR 0.1	-0.1426	-0.0370	0.0075	0.0456	0.1305	-0.0002
MCAR 0.2	-0.0857	-0.0233	0.0075	0.0291	0.1229	0.0032
MCAR 0.4	-0.0642	-0.0254	-0.0082	0.0181	0.0860	-0.0028
MCAR 0.8	-0.0587	-0.0204	-0.0015	0.0072	0.0570	-0.0036
MVN						
MAR 0.1	-0.4539	-0.1142	-0.0136	0.0701	0.2954	-0.0269
MAR 0.2	-0.2990	-0.0847	-0.0184	0.0558	0.2544	-0.0224
MAR 0.4	-0.2946	-0.1357	-0.0424	0.0344	0.3008	-0.0443
MAR 0.8	-0.3395	-0.1057	-0.0059	0.0821	0.3524	-0.0121
MCAR 0.1	-0.3533	-0.1295	-0.0078	0.0755	0.4474	-0.0175
MCAR 0.2	-0.4145	-0.0921	-0.0005	0.0883	0.3110	-0.0077
MCAR 0.4	-0.2794	-0.0979	-0.0066	0.0802	0.3103	-0.0075
MCAR 0.8	-0.2464	-0.0878	-0.0151	0.0688	0.2872	-0.0032

(continued)

Scenario	Min	Q1	Median	Q3	Max	Mean
Spatial MVN						
MAR 0.1	-0.4138	-0.1287	-0.0308	0.0571	0.2665	-0.0359
MAR 0.2	-0.4862	-0.1411	-0.0439	0.0257	0.3829	-0.0488
MAR 0.4	-0.3142	-0.1306	0.0037	0.1088	0.2775	-0.0128
MAR 0.8	-0.3158	-0.1080	-0.0235	0.0633	0.3090	-0.0232
MCAR 0.1	-0.3324	-0.0904	-0.0060	0.0707	0.4022	-0.0062
MCAR 0.2	-0.3288	-0.0706	-0.0007	0.0735	0.3007	0.0030
MCAR 0.4	-0.3425	-0.1339	-0.0408	0.0506	0.2084	-0.0394
MCAR 0.8	-0.4041	-0.1081	-0.0335	0.0536	0.3094	-0.0329

Table 7. Summary of parameter from Brunei-Muara complete and imputed data fitting process using JAGS

Parameter	**Complete Data***	**Missing Data***
Factor loadings		
λ_1	0.443 [0.385, 0.509]	0.433 [0.372, 0.499]
λ_2	0.799 [0.745, 0.848]	0.797 [0.745, 0.854]
λ_3	0.761 [0.713, 0.809]	0.770 [0.721, 0.817]
λ_4	0.782 [0.730, 0.834]	0.796 [0.743, 0.851]
Regression coefficients		
β_1	0.298 [0.275, 0.322]	0.314 [0.289, 0.340]
β_2	0.550 [0.520, 0.584]	0.561 [0.528, 0.595]
β_3	0.049 [0.022, 0.079]	0.034 [0.003, 0.064]
β_4	0.038 [0.008, 0.068]	0.028 [-0.004, 0.057]
Scaling factor to spatial influence strength		
κ_1	0.537 [0.377, 0.706]	0.543 [0.376, 0.706]
κ_2	0.471 [0.344, 0.613]	0.466 [0.346, 0.593]
κ_3	0.323 [0.186, 0.449]	0.327 [0.194, 0.465]
κ_4	0.443 [0.301, 0.590]	0.474 [0.323, 0.607]
Spatial autocorrelation		
ρ	0.292 [0.050, 0.698]	0.290 [0.050, 0.677]
Variance Components		
ν^2	0.155 [0.143, 0.166]	0.154 [0.142, 0.166]
τ^2	0.532 [0.199, 0.953]	0.529 [0.184, 0.931]
Variance of the independent noise		
θ_1	0.781 [0.726, 0.841]	0.781 [0.724, 0.844]
θ_2	0.319 [0.277, 0.362]	0.315 [0.274, 0.356]
θ_3	0.381 [0.341, 0.421]	0.370 [0.332, 0.408]
θ_4	0.349 [0.307, 0.389]	0.338 [0.300, 0.378]

Posterior Mean [95% CI].

References

Arnone, Eleonora, L. Sangalli, and A. Vicini. 2021. "Smoothing Spatio-Temporal Data with Complex Missing Data Patterns." *Statistical Modelling* 23: 327–56. https://doi.org/10.1177/1471082X211057959.

Baker, Jannah, Nicole White, and Kerrie Mengersen. 2014. "Missing in Space: An Evaluation of Imputation Methods for Missing Data in Spatial Analysis of Risk Factors for Type II Diabetes." *International Journal of Health Geographics* 13: 1–13.

Beesley, Lauren J., I. Bondarenko, Michael R. Elliot, A. Kurian, S. Katz, and Jeremy M. G. Taylor. 2021. "Multiple Imputation with Missing Data Indicators." *Statistical Methods in Medical Research* 30: 2685–2700. https://doi.org/10.1177/09622802211047346.

Bertsimas, D., C. Pawlowski, and Ying Daisy Zhuo. 2017. "From Predictive Methods to Missing Data Imputation: An Optimization Approach." *J. Mach. Learn. Res.* 18: 196:1–39.

Besag, Julian, Jeremy York, and Annie Mollié. 1991. "Bayesian Image Restoration, with Two Applications in Spatial Statistics." *Annals of the Institute of Statistical Mathematics* 43 (March): 1–20. https://doi.org/10.1007/BF00116466.

Brini, A., and Edwin van den Heuvel. 2023. "Missing Data Imputation with High-Dimensional Data." *The American Statistician* 78: 240–52. https://doi.org/10.1080/00031305.2023.2259962.

Chapon, Antoine, T. Ouarda, and Y. Hamdi. 2023. "Imputation of Missing Values in Environmental Time Series by d-Vine Copulas." *Weather and Climate Extremes*. https://doi.org/10.1016/j.wace.2023.100591.

Demirtas, H. 2018. "Flexible Imputation of Missing Data." *Journal of Statistical Software* 85: 1–5. https://doi.org/10.18637/JSS.V085.B04.

Donegan, Connor. 2021. "Building Spatial Conditional Autoregressive (CAR) Models in the Stan Programming Language."

Du, H., Craig K. Enders, B. Keller, T. Bradbury, and B. Karney. 2021. "A Bayesian Latent Variable Selection Model for Nonignorable Missingness." *Multivariate Behavioral Research* 57: 478–512. https://doi.org/10.1080/00273171.2021.1874259.

Dyk, D. V. van, and X. Meng. 2001. "The Art of Data Augmentation." *Journal of Computational and Graphical Statistics* 10: 1–50. https://doi.org/10.1198/10618600152418584.

Freni-Sterrantino, A., Massimo Ventrucci, and H. Rue. 2017. "A Note on Intrinsic Conditional Autoregressive Models for Disconnected Graphs." *Spatial and Spatio-Temporal Epidemiology* 26: 25–34. https://doi.org/10.1016/j.sste.2018.04.002.

Gelman, A. 2004. "Prior Distributions for Variance Parameters in Hierarchical Models (Comment on Article by Browne and Draper)." *Bayesian Analysis* 1: 515–34. https://doi.org/10.1214/06-BA117A.

Gong, Yongshun, Zhibin Li, Jian Zhang, Wei Liu, Yilong Yin, and Yu Zheng. 2023. "Missing Value Imputation for Multi-View Urban Statistical Data via Spatial Correlation Learning." *IEEE Transactions on Knowledge and Data Engineering* 35: 686–98. https://doi.org/10.1109/TKDE.2021.3072642.

Grund, S., O. Lüdtke, and A. Robitzsch. 2018. "Multiple Imputation of Missing Data for Multilevel Models." *Organizational Research Methods* 21: 111–49. https://doi.org/10.1177/1094428117703686.

Hill, J., A. Gelman, J. Kropko, and Ben Goodrich. 2014. "Multiple Imputation for Continuous and Categorical Data: Comparing Joint Multivariate Normal and Con-

ditional Approaches." *Political Analysis* 22: 497–519. https://doi.org/10.1093/pan/mpu007.

Jamil, Haziq. 2025. "A Spatio-Temporal Analysis of House Prices in Brunei Darussalam." *Quality & Quantity*, 1–32.

Jamil, Haziq, Amira Barizah Noorosmawie, Hafeezul Waezz Rabu, and Lutfi Abdul Razak. 2025. "From Archives to AI: Residential Property Data Across Three Decades in Brunei Darussalam." *Data in Brief*, March. https://doi.org/10.1016/j.dib.2025.111505.

Kinasih, Indira Puteri, Haziq Jamil, and Elvynna Leong. 2025. "Conditionally Autoregressive Models for House Price Data: Insights from a Comparative Simulation Study." *Journal of Statistical Theory and Applications*, 1–27.

Lai, M. H. 2021. "Adjusting for Measurement Noninvariance with Alignment in Growth Modeling." *Multivariate Behavioral Research* 58: 30–47. https://doi.org/10.1080/00273171.2021.1941730.

Lawson, A., C. Rotejanaprasert, P. Moraga, and J. Choi. 2015. "A Shared Neighbor Conditional Autoregressive Model for Small Area Spatial Data." *Environmetrics* 26: 383–92. https://doi.org/10.1002/env.2346.

Lee, Duncan. 2013. "CARBayes: An r Package for Bayesian Spatial Modeling with Conditional Autoregressive Priors." *Journal of Statistical Software* 55 (13): 1–24. https://doi.org/10.18637/jss.v055.i13.

Lee, Duncan, A. Rushworth, and Gary Napier. 2018. "Spatio-Temporal Areal Unit Modeling in r with Conditional Autoregressive Priors Using the CARBayesST Package." *Journal of Statistical Software* 84: 1–39. https://doi.org/10.18637/JSS.V084.I09.

Lee, Katherine J., and J. Carlin. 2010. "Multiple Imputation for Missing Data: Fully Conditional Specification Versus Multivariate Normal Imputation." *American Journal of Epidemiology* 171 5: 624–32. https://doi.org/10.1093/aje/kwp425.

Lee, Min Cherng, and R. Mitra. 2016. "Multiply Imputing Missing Values in Data Sets with Mixed Measurement Scales Using a Sequence of Generalised Linear Models." *Comput. Stat. Data Anal.* 95: 24–38. https://doi.org/10.1016/j.csda.2015.08.004.

Leitgöb, H., Daniel Seddig, T. Asparouhov, Dorothée Behr, E. Davidov, Kim De Roover, S. Jak, et al. 2022. "Measurement Invariance in the Social Sciences: Historical Development, Methodological Challenges, State of the Art, and Future Perspectives." *Social Science Research* 110: 102805. https://doi.org/10.1016/j.ssresearch.2022.102805.

Li, Huiping, Meng Li, Xi Lin, Fang He, Yinhai Wang, and Yinhai Wang. 2020. "A Spatiotemporal Approach for Traffic Data Imputation with Complicated Missing Patterns." *Transportation Research Part C-Emerging Technologies* 119: 102730. https://doi.org/10.1016/j.trc.2020.102730.

Lin, Wanyu, Sumei Wang, Yu Yang, Youwu Wang, Qianyi Chen, and Jiannong Cao. 2024. "Multistage Graph Convolutional Network with Spatial Attention for Multivariate Time Series Imputation." *IEEE Transactions on Neural Networks and Learning Systems* PP. https://doi.org/10.1109/TNNLS.2024.3486349.

Liu, Jin, Jing Zhou, Wei Lan, and Hansheng Wang. 2023. "Spatial Dynamic Panel Models with Missing Data." *Stat* 12. https://doi.org/10.1002/sta4.585.

Ma, Zhihua, Guanyu Hu, and Ming-Hui Chen. 2020. "Bayesian Hierarchical Spatial Regression Models for Spatial Data in the Presence of Missing Covariates with Applications." *Applied Stochastic Models in Business and Industry*. https://doi.org/10.1002/asmb.2568.

Miao, Xiaoye, Yangyang Wu, Lu Chen, Yunjun Gao, and Jianwei Yin. 2023. "An Experimental Survey of Missing Data Imputation Algorithms." *IEEE Transactions on Knowledge and Data Engineering* 35: 6630–50. https://doi.org/10.1109/TKDE.2022.

3186498.

Nguyen, C., J. Carlin, and Katherine J. Lee. 2021. "Practical Strategies for Handling Breakdown of Multiple Imputation Procedures." *Emerging Themes in Epidemiology* 18. https://doi.org/10.1186/s12982-021-00095-3.

Paik, M. 1997. "The Generalized Estimating Equation Approach When Data Are Not Missing Completely at Random." *Journal of the American Statistical Association* 92: 1320–29. https://doi.org/10.1080/01621459.1997.10473653.

Skarstein, Emma, S. Martino, and S. Muff. 2023. "A Joint Bayesian Framework for Missing Data and Measurement Error Using Integrated Nested Laplace Approximations." *Biometrical Journal* 65. https://doi.org/10.1002/bimj.202300078.

Tanner, M., and W. Wong. 1987. "The Calculation of Posterior Distributions by Data Augmentation." *Journal of the American Statistical Association* 82: 528–40. https://doi.org/10.1080/01621459.1987.10478458.

Tian, Maozai, and Jinwen Liang. 2024. "Imputed Quantile Vector Autoregressive Model for Multivariate Spatial–Temporal Data." *Statistical Analysis and Data Mining: The ASA Data Science Journal* 17. https://doi.org/10.1002/sam.11658.

Van Buuren, Stef, and Karin Groothuis-Oudshoorn. 2011. "Mice: Multivariate Imputation by Chained Equations in r." *Journal of Statistical Software* 45: 1–67.

Vermunt, Jeroen K, and Jay Magidson. 2021. "How to Perform Three-Step Latent Class Analysis in the Presence of Measurement Non-Invariance or Differential Item Functioning." *Structural Equation Modeling: A Multidisciplinary Journal* 28 (3): 356–64.

Wang, Wei, and Lung-fei Lee. 2013. "Estimation of Spatial Panel Data Models with Randomly Missing Data in the Dependent Variable." *Regional Science and Urban Economics* 43: 521–38. https://doi.org/10.1016/J.REGSCIURBECO.2013.02.001.

Wang, Zhiqiang, and Ni Tang. 2020. "Bayesian Quantile Regression with Mixed Discrete and Nonignorable Missing Covariates." *Bayesian Analysis*. https://doi.org/10.1214/19-BA1165.

Zahmatkesh, Samira, and M. Mohammadzadeh. 2020. "Bayesian Prediction of Spatial Data with Non-Ignorable Missingness." *Statistical Papers*, 1–22. https://doi.org/10.1007/s00362-020-01186-0.

Zhang, Qiang, Qiangqiang Yuan, Chao Zeng, Xinghua Li, and Yancong Wei. 2018. "Missing Data Reconstruction in Remote Sensing Image with a Unified Spatial–Temporal–Spectral Deep Convolutional Neural Network." *IEEE Transactions on Geoscience and Remote Sensing* 56: 4274–88. https://doi.org/10.1109/TGRS.2018.2810208.